

QC meet CQ: Quantum Conjunctive Queries

Floris Geerts

University of Antwerp

Antwerpen, Belgium

floris.geerts@uantwerpen.be

Rihan Hai

University of Delft

Delft, Netherlands

R.Hai@tudelft.nl

Abstract

We explore how recent methods for evaluating conjunctive queries (CQs) can help to efficiently simulate quantum circuits (QCs), i.e., computing output amplitudes from a given input state.

ACM Reference Format:

Floris Geerts and Rihan Hai. 2025. QC meet CQ: Quantum Conjunctive Queries. In *Workshop on Quantum Computing and Quantum-Inspired Technology for Data-Intensive Systems and Applications (Q-Data '25)*, June 22–27, 2025, Berlin, Germany. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3736393.3736696>

Intro. Recent work suggests that relational database systems are well-suited for quantum circuit (QC) simulation, thanks to their portability [2] and out-of-core processing capabilities [7]. These advantages are particularly relevant when simulations rely on sparse tensor computations [5]. QC simulation is often modeled as tensor network manipulation, where the circuit's structure forms a graph. This representation is central to analyzing tractability. A seminal result by Markov and Shi [6] shows that a QC with T gates and a graph of treewidth d can be simulated in time $T^{O(1)}e^{O(d)}$ —polynomial in the number of gates, but exponential in treewidth, reflecting entanglement complexity. Their analysis assumes dense tensor representation, storing all entries, including zeros. *Our goal is to better understand QC simulation complexity in the sparse setting, where only non-zero entries are stored.* We approach this by (i) mapping QC simulation to conjunctive query (CQ) evaluation, and (ii) leveraging CQ complexity analysis to study QC simulation.

Quantum simulation=Query evaluation. The first challenge is converting QCs into CQs. Prior work performs the required tensor contractions—essentially extended matrix multiplications—using SQL queries with aggregation (e.g., SUM, GROUP BY) over large relations [2, 5, 7]. Though theoretically complex, these aggregate queries can be reformulated as Select–Project–Join queries, or Conjunctive Queries [1]. The key is to evaluate CQs over semiring-annotated relations: with complex numbers as the semiring, relations represent tensors, joins implement element-wise multiplication, and projections perform summation over contracted indices. This QC–CQ connection has been observed before [3], and general tensor computations can be encoded similarly. We formally define QCs and provide a general translation to CQs, which we call *quantum CQs*.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Q-Data '25, Berlin, Germany

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1944-8/25/06

<https://doi.org/10.1145/3736393.3736696>

Hypertree width of quantum conjunctive queries. An initial observation is that the treewidth of a quantum CQ [4] aligns with the treewidth of the corresponding QC [6]. Treewidth is defined via the CQ's primal graph, where nodes represent variables and edges connect variables co-occurring in a relation. In QC terms, variables map to qubits and relations to gates, making the primal graph of a quantum CQ the dual of the QC's circuit graph. However, graph-based representations are not always ideal. For instance, acyclic CQs can have arbitrarily large treewidth, despite being evaluable in linear time via the Yannakakis algorithm. To address this, Gottlob et al. [4] introduced hypergraphs, which generalize graphs by allowing hyperedges—connecting multiple nodes. In this view, variables (qubits) are nodes, and relations (gates) are hyperedges, as illustrated below for the three-qubit GHZ state.

What does this yield? By considering a hypertree decomposition of the hypergraph, one obtains a structural measure called *hypertree width* [4]. If a CQ has hypertree width d , it can be evaluated in time $O(m \log(r)r^d)$, where r is the size of the largest relation and m the size of the decomposition tree. In the quantum setting, r corresponds to *gate sparsity*, i.e., the maximum number of non-zero entries in any gate tensor. Notably, if the quantum CQ has an acyclic hypergraph, then $d = 1$ and the evaluation complexity is $O(mr)$. We want to conclude with our key takeaway: refined hypergraph measures allow a deeper understanding of quantum simulation complexity in terms of gate sparsity.

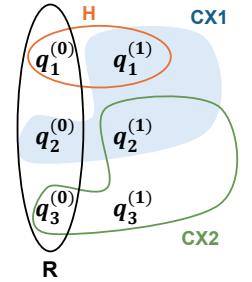


Figure 1: Hypergraph of 3-qubit GHZ state preparation circuit of (binary) Hadamard and (ternary) CNOT gates. Here, R denotes the initial state.

References

- [1] Mahmoud Abo Khamis, Hung Q. Ngo, and Atri Rudra. 2016. FAQ: Questions Asked Frequently. In *PODS*. 13–28. <https://doi.org/10.1145/2902251.2902280>
- [2] Mark Blacher, Julien Klaus, Christoph Staudt, Sören Laue, Viktor Leis, and Joachim Giesen. 2023. Efficient and Portable Einstein Summation in SQL. *Proc. ACM Manag. Data* 1, 2, Article 121 (2023). <https://doi.org/10.1145/3589266>
- [3] Jeffrey M. Dudek, Leonardo Dueñas-Osorio, and Moshe Y. Vardi. 2019. Efficient Contraction of Large Tensor Networks for Weighted Model Counting through Graph Decompositions. *CoRR* (2019). <http://arxiv.org/abs/1908.04381>
- [4] Georg Gottlob, Nicola Leone, and Francesco Scarcello. 1999. Hypertree decompositions and tractable queries. In *PODS*. 21–32. <https://doi.org/10.1145/303976.303979>
- [5] Rihan Hai, Shih-Han Hung, Tim Coopmans, Tim Littau, and Floris Geerts. 2025. Data Management in the Noisy Intermediate-Scale Quantum Era. 18 (2025), 1720–1729. Issue 6. <https://doi.org/10.14778/3725688.3725701>
- [6] Igor L. Markov and Yaoyun Shi. 2008. Simulating Quantum Computation by Contracting Tensor Networks. *SICOMP* 38, 3 (2008), 963–981. <https://doi.org/10.1137/050644756>
- [7] Immanuel Trummer. 2024. Towards Out-of-Core Simulators for Quantum Computing. In *Q-Data*. <https://doi.org/10.1145/3665225.3665441>